

Snowflake Customer Intelligence & Marketing Optimisation Platform

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Executive Summary

The objective is to build a production-style data platform using Snowflake's retail sample datasets.

The solution simulates how a modern enterprise transforms data into trusted data products, applies governance and quality controls, and delivers business value through analytics and AI.

The final solution should allow business users to understand:

- Who their customers are
- Which customers generate the most value
- Which customers are likely to return products
- Which promotions are effective
- Which regions perform best
- Which customer segments should receive future marketing
- Why trends are occurring
- Answers to natural language questions via Cortex

Source Data

Customer Domain

- CUSTOMER
- CUSTOMER_ADDRESS
- CUSTOMER_DEMOGRAPHICS
- HOUSEHOLD_DEMOGRAPHICS
- INCOME_BAND

Sales Domain -

- WEB_SALES
- STORE_SALES
- CATALOG_SALES

Returns Domain

- WEB_RETURNS

- STORE_RETURNS
- CATALOG_RETURNS

Marketing Domain

- PROMOTION
- WEB_PAGE
- WEB_SITE

Product Domain

- ITEM
- INVENTORY
- WAREHOUSE

Target Architecture

Source Tables

- > RAW Layer
- > CURATED Layer
- > MODELLED Layer ->
CONSUMER Layer
- > Semantic Views
- > Cortex Search / Cortex Agent

Phase 1 – Ingestion

- Load using COPY INTO
- Capture metadata including LOAD_ID, LOAD_DATE, SOURCE_FILE_NAME and BATCH_ID

Phase 2 – Raw Layer

- Store source data exactly as received
- No transformations
- Preserve historical loads

Phase 3 – Curated Layer

- Standardise data types
- Handle nulls and duplicates
- Cleanse invalid records
- Derive age bands, regions and classifications

Phase 4 – Modelled Layer

Fact Tables:

- FACT_SALES
- FACT_RETURNS
- FACT_CUSTOMER_ACTIVITY

- FACT_PROMOTION_RESPONSE

Dimension Tables:

- DIM_CUSTOMER
- DIM_PRODUCT
- DIM_DATE
- DIM_STORE
- DIM_REGION
- DIM_PROMOTION

Phase 5 – Consumer Layer

Data Product 1: Customer 360

- Lifetime Spend
- Average Order Value
- Purchase Frequency
- Recency
- Return Rate
- Promotion Response

Data Product 2: Returns Intelligence

- Return Percentage
- Return Frequency
- Return Value
- Days To Return

Data Product 3: Campaign Effectiveness

- Revenue Uplift
- Conversion Rate
- Promotion Revenue
- ROI Analysis

Data Product 4: Customer Value Segmentation

- Platinum
- Gold
- Silver
- Bronze
- At Risk Customers
- Dormant Customers

Phase 6 – Testing Technical

Tests:

- Row counts
- Duplicate checks

- Null checks
- Column validation

Business Tests:

- Sales values > 0
- Return dates after sale dates
- Customer exists
- Revenue reconciles

Phase 7 – Data Quality

- Completeness
- Freshness
- Validity
- Referential Integrity

Phase 8 – Dashboards

- Customer Dashboard
- Sales Dashboard
- Returns Dashboard
- Marketing Dashboard

Phase 9 – AI & Semantic Layer Semantic Views:

- Customer Lifetime Value
- Return Rate
- Average Basket
- Promotion Conversion Rate
- Sales Uplift

Cortex Search:

- Product Reviews
- Campaign Documents
- Regional Performance Reports

Example Questions:

- Which customer segment generates the highest revenue?
- Who are the 10 highest value customers?
- Which regions have the most sales?

Success Criteria

- Automated testing
- Data Quality monitoring
- Full layered architecture

- Customer 360 data product
- Dashboards
- Semantic Views
- Cortex Search
- Cortex Agent